

Millisecond-Scale Neural Operator Surrogates for Double-Null Free-Boundary Grad–Shafranov Equilibria

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The Grad–Shafranov (GS) equation is a key relation governing the ideal magnetohydrodynamic (MHD) equilibrium in tokamak plasmas. One of the major challenges in using free-boundary GS solvers in real-time control workflows and large-scale optimization and design scenarios is the large computational cost and sample-dependent latency of nonlinear Picard iteration. In particular, repeated high-fidelity evaluations for diverted tokamak equilibria can become prohibitive when the equilibrium computation is inside an optimization, modeling, or control-oriented loop. Here we train a geometrically conditioned Fourier Neural Operator (FNO) to learn a computationally efficient representation of a constrained forward map from spatial coordinates, scalar operating parameters ($P_{\text{axis}}, I_p, f_{\text{vac}}$), and prescribed X-point locations to the full poloidal-flux field $\psi(R, Z)$. The FNO model is trained on a controlled family of constrained double-null free-boundary equilibrium configurations generated with the FREEGS solver for a single fixed machine geometry and prescribed topology. The best model achieves a mean relative L^2 error of 0.05%, with the test error following an empirical $N^{-0.68}$ power law over the range $N_{\text{train}} \in \{500, 1000, 2000, 5000\}$, indicating systematic improvement with training-set size over the tested regime. It also recovers both X-points to within 0.2 cm and localizes the O-point to 0.03 cm. As a physics-consistency diagnostic, the predicted fields satisfy an external finite-difference GS residual evaluation at the same level as the ground-truth fields, with a mean normalized residual of 2.29, indistinguishable from the 2.29 ± 0.06 baseline obtained on FREEGS solutions using the same diagnostic. The trained FNO surrogate evaluates a single equilibrium in 2.77 ms on GPU and 25.6 ms on CPU, corresponding to speedups of $\sim 640\times$ and $\sim 69\times$ relative to FREEGS as configured in these experiments, with near-deterministic latency (p95/median = 1.01). These results demonstrate that neural-operator surrogates can provide accurate, geometrically precise, and millisecond-scale equilibrium evaluations for magnetic-confinement fusion workflows within a prescribed topology and machine geometry.

I. INTRODUCTION

Plasma equilibrium in axisymmetric geometry in fusion devices is governed by the nonlinear elliptic Grad–Shafranov (GS) equation for the poloidal magnetic flux $\psi(R, Z)^{1-3}$. The resulting field determines the magnetic axis, flux-surface geometry, separatrix shape, and X-point structure, and accordingly it is critical for tasks such as stability analysis, transport modeling, equilibrium reconstruction, and plasma shape control⁴. Accurate and rapid evaluation of the plasma equilibrium is therefore a central computational task in magnetic-confinement fusion.

A key challenge in free-boundary equilibrium modeling is that the plasma boundary is not known in advance. In diverted configurations, it must emerge self-consistently from the plasma profiles, external coil fields, and global geometric constraints. Therefore production codes compute nonlinear GS updates while enforcing constraints, including isoflux conditions and explicit X-point locations⁵⁻⁷. These approaches are robust and mature, but they employ iterative nonlinear numerical algorithms with sample-dependent convergence latency, and since equilibria are evaluated many times, have non-negligible computational cost. Specifically, repeated high-fidelity GS evaluations can become prohibitive in many-query settings such as uncertainty quantification, outer-loop optimization, integrated modeling, and control-oriented workflows⁸, while recent surrogate-equilibrium efforts such

as EFIT-Prime (Equilibrium FIT) highlight the complementary need for fast, uncertainty-aware equilibrium reconstruction in real-time and control-relevant settings⁹. Real-time equilibrium reconstruction has been achieved in experimental settings through machine-specific optimization^{6,7}, but such systems are typically engineered around the diagnostics, actuators, and temporal coherence of a specific device. On the other hand, emerging use cases such as digital twins, controller training, and rapid parametric scans require repeated evaluation across families of operating points for which warm-started iteration is either unavailable or impractical.

Machine learning (ML) has become an important tool in nuclear fusion research, with applications ranging from transport modeling and integrated simulation to disruption prediction and plasma control¹⁰⁻³¹. In particular, ML approaches have been applied to GS equilibria in several recent works. Physics-informed neural networks (PINNs)³² have been used to solve the GS equation³³, but a trained PINN model typically evaluates a single problem instance or a narrow family rather than providing solutions across a wide parametric set. Neural-network and reduced-order approaches have also been developed for fast full-field equilibrium reconstruction from magnetic diagnostics. Joungh et al. developed a diagnostic-constrained neural-network Grad–Shafranov solver for KSTAR³⁴ that maps measured magnetic signals and spatial coordinates to $\psi(R, Z)$ and $\Delta^* \psi$, reproducing off-line EFIT-quality equilibria with approximately mil-

lisecond latency²⁷. Lu et al. similarly developed a deep-learning reconstruction model for EAST³⁵ that maps magnetic measurements, coil currents, and plasma current to the two-dimensional poloidal flux $\Psi(R, Z)$ trained on off-line EFIT reconstructions²⁹. GS-DeepNet further introduced a self-teaching physics-constrained framework for KSTAR equilibrium reconstruction based on neural representations of the Grad-Shafranov equation²⁸. Wai et al. developed Eqnet for NSTX-U³⁶, a neural-network equilibrium model trained on EFIT01 reconstructions that can operate either in a diagnostic-reconstruction mode or in a forward mode using coil currents, vessel currents, and pressure and current-profile information to predict equilibrium flux surfaces³⁰. More recently, McClenaghan et al. augmented machine-learning equilibrium reconstruction with EFIT Green's functions by representing the toroidal current density in a PCA (principal component analysis) basis and constructing consistent basis functions for ψ and synthetic magnetic diagnostics³¹. These works demonstrate the potential of ML for rapid equilibrium reconstruction and fast surrogate modeling. However, most of them address the inverse problem of reconstruction from measurements, while the forward Eqnet model provides a closely related fully connected neural-network surrogate for EFIT-based free-boundary flux-surface prediction. Reinforcement-learning approaches have produced striking demonstrations of magnetic control at TCV³⁷ and KSTAR^{12,13}. Such approaches, and control-optimization methods more generally, benefit from fast and repeatedly queryable inner-loop models whose latency is both low and predictable.

Operator learning provides a natural framework for this requirement. Rather than solving one partial differential equation (PDE) instance at a time, a neural operator learns the map between families of inputs and their corresponding solution fields^{38–40}. Once trained, the same model can be applied to new members of the learned family at the cost of a single forward evaluation. DeepONet established a practical architecture for nonlinear operator approximation⁴¹, while the Fourier Neural Operator (FNO) introduced spectral convolutions that have proven effective across broad classes of parametric PDEs⁴². The FNO architecture is particularly well suited for GS equilibria since elliptic problems are globally coupled. A local change in current, pressure, or boundary geometry redistributes flux throughout the computational domain, and spectral convolutions provide an efficient mechanism for capturing such long-range interactions. FNOs have shown strong performance in physics applications^{43,44} and in plasma physics for reduced turbulence models and MHD-related tasks^{25,26}.

The constrained free-boundary regime relevant to diverted tokamaks remains relatively unexplored from this operator-learning perspective. In this regime, X-point location, separatrix topology, and isoflux consistency must be preserved simultaneously. Prior neural surrogates for GS-adjacent tasks have focused primarily on fixed-boundary formulations, diagnostic reconstruction from measurements, EFIT-based forward flux-surface prediction, or turbulence and MHD dynamics rather than geometry-conditioned neural-operator surrogates for constrained free-boundary equilibrium

generation^{25,26,28–30,33}.

In this work, we use a geometry-conditioned FNO to learn a computationally efficient representation of a constrained forward map from spatial coordinates, scalar operating parameters, and prescribed X-point coordinates to the full poloidal-flux field. This approach differs from diagnostic reconstruction models, which infer equilibria from measurements, and from actuator-conditioned forward models such as Eqnet, which use coil, vessel, and profile information to predict EFIT-based flux surfaces. The learned FNO here should therefore be regarded as a surrogate for a constrained free-boundary design problem, not as an unconstrained actuator-to-equilibrium map. In particular, we train the model on a controlled family of constrained double-null free-boundary equilibria generated with the FREEGS code⁴⁵ for a single fixed machine geometry and prescribed topology. The explicit conditioning on X-point coordinates is a deliberate design choice – it anchors the divertor topology directly and is appropriate for design and control workflows in which the target X-point geometry is prescribed or measurable.

We assess the trained FNO model using both field-level and geometry-aware diagnostics. Specifically, in addition to relative L^2 error, we evaluate separatrix shape, X- and O-point localization, enclosed separatrix area, and boundary-flux consistency. We also test whether the predicted fields satisfy an external finite-difference GS residual diagnostic at the same level as the ground-truth FREEGS fields, using the same diagnostic on both predicted and reference fields. These results demonstrate that, within the prescribed topology and machine geometry studied in this work, neural operators can provide accurate, geometrically faithful, and millisecond-scale surrogates for repeated constrained free-boundary equilibrium evaluation.

The paper is organized as follows. After the introductory discussion in this section, Section II describes the theoretical framework, data generation procedure, geometry-conditioned FNO architecture and training, and evaluation metrics. Section III presents our results, namely the data-scaling study, field-level accuracy, geometry-aware validation, inference-latency benchmarks, and finite-difference GS residual diagnostic. Section IV discusses the implications for control and integrated modeling, interprets the observed error structure, and summarizes the main limitations and future extensions. We conclude in Section V with a summary of the main results and a brief outlook.

II. METHODS

A. Free-boundary GS formulation and data generation

In axisymmetric geometry, ideal MHD equilibrium satisfies the Grad-Shafranov equation for the poloidal magnetic flux $\psi(R, Z)$ ^{1,3}:

$$\Delta^* \psi(R, Z) = -\mu_0 R^2 \frac{dp}{d\psi} - \frac{1}{2} \frac{dF^2}{d\psi}, \quad (1)$$

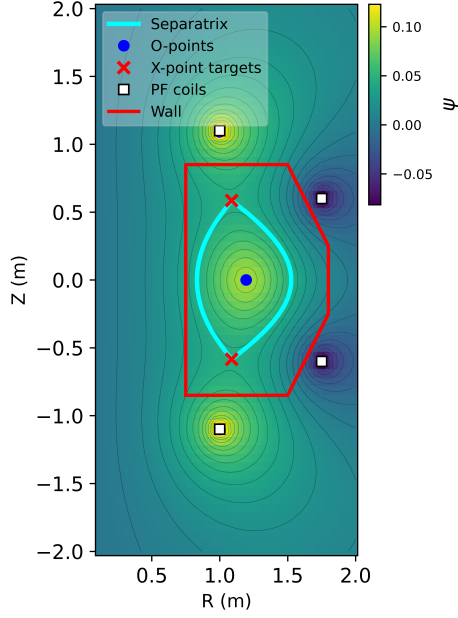


FIG. 1. Representative double-null free-boundary equilibrium generated with FREEGS for input parameters $P_{\text{axis}} = 1200$ Pa, $I_p = 2.0 \times 10^5$ A, $f_{\text{vac}} = 1.75$, with X-point targets at $(R, Z) = (1.1, +0.6)$ m (upper) and $(1.1, -0.6)$ m (lower). The cyan contour shows the separatrix $\psi = \psi_{\text{bdry}}$, the filled blue circle is the magnetic O-point, red crosses mark the X-point target locations, the red polygon is the wall boundary, and open squares denote PF coil positions. This configuration is representative of the training distribution.

where $\Delta^* = R\partial_R(R^{-1}\partial_R) + \partial_Z^2$, $p(\psi)$ is the plasma pressure, and $F(\psi) = RB_\phi$ denotes the toroidal-field function. For a free-boundary equilibrium, the plasma boundary is not prescribed a priori. Instead, it must be determined self-consistently together with the surrounding coil response and the plasma profiles.

In this study, we consider diverted double-null equilibrium configurations constrained by prescribed upper and lower X-point locations together with an isoflux condition. The boundary flux is defined as $\psi_{\text{bdry}} = \frac{1}{2}(\psi_X^{\text{lo}} + \psi_X^{\text{up}})$, where the superscripts denote the lower and upper X-points. Each baseline equilibrium evaluation is therefore the solution of a nonlinear elliptic PDE coupled to global constraint updates. In practice, the problem is solved with Picard iteration while enforcing the X-point and isoflux conditions at each step. The surrogate developed below should therefore be interpreted as a fast approximation to this constrained free-boundary design problem, not as an unconstrained actuator-to-equilibrium map.

All data were generated using FREEGS⁴⁵, an open-source free-boundary Grad-Shafranov solver, for the fixed FREEGS TestTokamak geometry. The equilibria were computed over the radial domain $R \in [0.1, 2.0]$ m and vertical range $Z \in [-2.0, 2.0]$ m on a uniform 65×65 grid. The pressure and toroidal-field profile shapes are fixed by the FREEGS setup used here, while their amplitudes and the global equilibrium are varied through scalar operating parameters. Thus, the dataset represents a controlled free-boundary equilibrium

family for a single machine geometry and prescribed double-null topology, rather than a cross-device or cross-topology distribution.

For each sample, the scalar operating parameters $(P_{\text{axis}}, I_p, f_{\text{vac}})$ are drawn uniformly from

$$P_{\text{axis}} \in [200, 3000] \text{ Pa}, \quad (2)$$

$$I_p \in [5 \times 10^4, 4 \times 10^5] \text{ A}, \quad (3)$$

$$f_{\text{vac}} \in [0.5, 3.0]. \quad (4)$$

The reference X-point targets are $(R_X, Z_X) = (1.1, \pm 0.6)$ m. Symmetric geometric jitter $|\delta R|, |\delta Z| \leq 0.02$ m is applied to the upper and lower nulls in a manner that preserves exact up-down symmetry. This broadens the separatrix family while preserving the double-null topology.

A base training pool of 5000 equilibria is generated with random seed 123. The scaling study uses nested subsets $N_{\text{train}} \in \{500, 1000, 2000, 5000\}$ drawn from this pool with a fixed permutation, using seed 12345, so that larger training sets strictly contain the smaller ones. Validation and test datasets, each containing 500 samples, are generated independently using seeds 456 and 789, respectively, and are held fixed across all experiments. During generation, each candidate solution is retained only if FREEGS converges, the solved plasma current satisfies $|I_{p,\text{sol}} - I_{p,\text{tgt}}|/|I_{p,\text{tgt}}| \leq 10\%$, and `find_critical` returns at least two distinct X-points on the resulting field. All 6000 candidate equilibrium solutions across the three datasets were accepted, giving a 100% acceptance rate. The learned distribution is therefore identical to the sampled prior, with no implicit parameter-space narrowing from filtering. A representative equilibrium from the training distribution is shown in Fig. 1.

B. Geometry-conditioned FNO surrogate

We approximate the parametric solution operator

$$\mathcal{G} : (R, Z, P_{\text{axis}}, I_p, f_{\text{vac}}, R_X^{\text{lo}}, Z_X^{\text{lo}}, R_X^{\text{up}}, Z_X^{\text{up}}) \mapsto \psi(R, Z), \quad (5)$$

which takes nine gridded input channels and maps them to the full two-dimensional flux field on the 65×65 computational grid. The first two channels are the spatial coordinates (R, Z) , which provide explicit geometric context. The next three channels are the scalar operating parameters $(P_{\text{axis}}, I_p, f_{\text{vac}})$, broadcast over the grid. Here P_{axis} is the plasma pressure at the axis, I_p is the plasma current, and f_{vac} controls the vacuum toroidal-field contribution in the FREEGS setup. The final four channels are the lower and upper X-point coordinates, also broadcast over the grid, which specify the double-null divertor geometry directly.

This geometry conditioning is deliberate. Rather than asking the network to infer the separatrix topology and saddle-point placement from the scalar operating parameters alone, we supply the X-point geometry explicitly as part of the operator input. In effect, the learned map is conditioned both on the operating point and on the divertor configuration to be reproduced. This choice is appropriate for constrained design

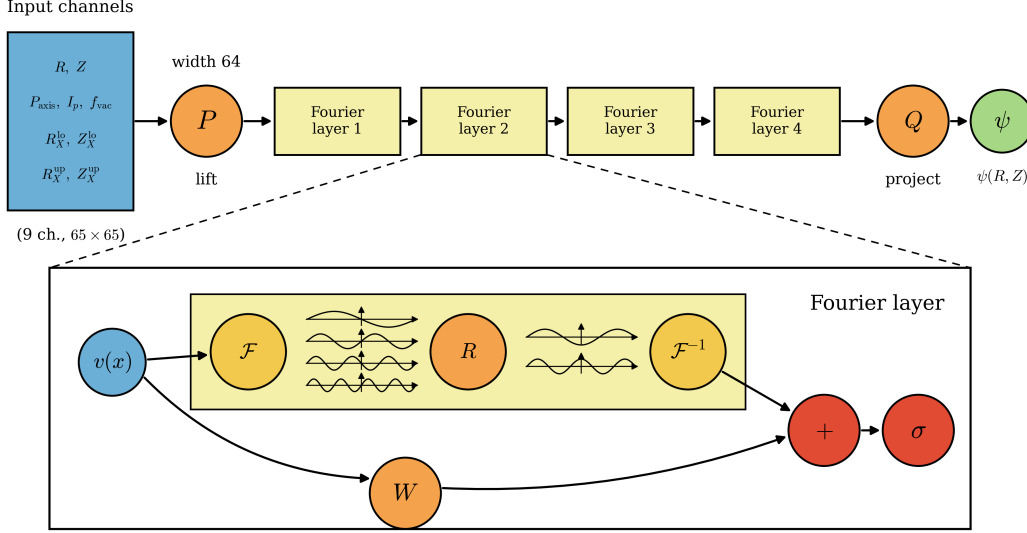


FIG. 2. Schematic of the geometry-conditioned Fourier Neural Operator used in this work. The input consists of nine channels on the computational grid: spatial coordinates (R, Z) , broadcast scalar operating parameters $(P_{\text{axis}}, I_p, f_{\text{vac}})$, and broadcast lower and upper X-point coordinates. A lifting layer maps these channels to a hidden representation, followed by four Fourier layers with truncated spectral convolutions and pointwise mixing. The final projection produces the full poloidal-flux field $\psi(R, Z)$ on a 65×65 grid. Geometry-aware diagnostics, including separatrix shape, X-/O-point localization, and the finite-difference Grad-Shafranov residual, are computed from the predicted field but are not used as training targets.

and control workflows in which the target X-point geometry is prescribed or measurable, but it should not be confused with an actuator-level forward model in which coil currents and profile parameters alone determine the resulting X-point locations.

We employ a two-dimensional Fourier Neural Operator (FNO)^{40,42}. For an input field $v(\mathbf{x})$, each operator layer applies a truncated spectral convolution,

$$\mathcal{H}v(\mathbf{x}) = \mathcal{F}^{-1}(R(\mathbf{k}) \mathcal{F}[v](\mathbf{k}))(\mathbf{x}), \quad (6)$$

where $\mathcal{F}$ denotes the discrete Fourier transform and $R(\mathbf{k})$ is a learned complex-valued weight tensor restricted to the lowest retained Fourier modes. Compared with purely local convolutional kernels, this spectral representation provides efficient global mixing, which is well matched to elliptic PDEs whose solutions depend on nonlocal coupling across the domain.

The network contains four Fourier layers with $n_{\text{modes}} = (16, 16)$ retained modes per dimension and hidden channel width 64, for a total of 4,770,241 trainable parameters. The model is implemented using the `neuraloperator` Python library³⁸. Spatial coordinates are passed explicitly as input channels – no additional positional embedding is used. A final pointwise linear projection maps the hidden representation to the single output channel $\psi(R, Z)$. The resulting geometry-conditioned FNO architecture is summarized schematically in Fig. 2.

C. Normalization and training

The scalar inputs $(P_{\text{axis}}, I_p, f_{\text{vac}})$ are standardized using the training-set mean and standard deviation. The four X-point coordinate channels are standardized in the same way. The spatial coordinates (R, Z) are linearly mapped to $[-1, 1]$. The target flux field ψ is normalized using the training-set mean and standard deviation. The training loss and reported relative L^2 errors are computed in this normalized representation. Physical-unit quantities, including the RMSE in Wb and all geometry-aware metrics, are computed after inverse transformation. This convention ensures that errors are evaluated consistently across the scaling study, while the physical RMSE provides a direct dimensional measure of the flux-field error.

Models are trained with the AdamW optimizer⁴⁶, using an initial learning rate of 10^{-3} and weight decay 10^{-4} . The training objective is the mean squared error between predicted and target flux fields in normalized units. A `ReduceLROnPlateau` scheduler halves the learning rate whenever the validation relative L^2 error fails to improve for 20 consecutive epochs, subject to a minimum learning rate of 10^{-5} . Early stopping is applied with a patience of 75 epochs, again based on the validation relative L^2 error. Mini-batches contain 16 samples, and training uses 4 data-loader workers.

For each training-set size in the scaling study, three independent random seeds are used for weight initialization. The dataset subsets themselves are kept fixed across seeds so that the reported variability reflects model initialization rather than data resampling.

D. Evaluation metrics and timing

Field-level accuracy is quantified by the relative L^2 error,

$$\epsilon_{\text{rel}} = \frac{\|\psi_{\text{pred}} - \psi_{\text{true}}\|_2}{\|\psi_{\text{true}}\|_2}, \quad (7)$$

and by the physical RMSE in Wb. All relative L^2 errors reported in this paper are computed in normalized units and then converted to percentages. Physical RMSE values are computed after inverse-normalizing both the predicted and target flux fields, so that they provide a direct dimensional measure of the flux-field error.

Geometry-aware metrics are evaluated for the best $N = 5000$ model over all 500 test samples. For each equilibrium, critical points of the predicted and ground-truth fields are located with the `find_critical` function from FREEGS. For each field, the separatrix is extracted as the largest closed contour of ψ at the mean X-point flux value,

$$\psi_{\text{bndry}} = \frac{1}{2} (\psi_X^{\text{lo}} + \psi_X^{\text{up}}). \quad (8)$$

To avoid contaminating closed-separatrix metrics with open divertor-leg contour artifacts, contour points whose radial distance from the contour centroid exceeds $Q_3 + 3 \times \text{IQR}$ of the contour-point radial distances are discarded. This post-processing is applied identically to the true and predicted contours before distance metrics are computed. We report the mean closest-point distance and Hausdorff distance between the predicted and true separatrices, the relative error in the enclosed separatrix area, the X-point location errors,

$$\Delta_X^{\text{lo,up}} = \sqrt{(R_X^{\text{pred}} - R_X^{\text{true}})^2 + (Z_X^{\text{pred}} - Z_X^{\text{true}})^2}, \quad (9)$$

the magnetic O-point location error, and the absolute error in ψ_{bndry} .

A Tukey-fence analysis, using the threshold $Q_3 + 1.5 \times \text{IQR}$, is used to characterize the heavy tail of the X-point error distribution. On the 500 held-out test predictions, `find_critical` returns two distinct X-points for every sample, for both the ground-truth FREEGS fields and the FNO predictions; there are no detection failures. Nevertheless, 5.8% of samples, corresponding to 29/500 cases, lie beyond the Tukey threshold for at least one of the two nulls. These geometric outliers are retained in all reported statistics and are the dominant contributor to the heavy Hausdorff tail discussed in Sec. III C; they are not a distinct failure mode, but rather the upper tail of the field-level error distribution amplified geometrically at the saddle points.

Inference latency is measured at batch size 1 with synchronized CUDA events on an NVIDIA A100 SXM4-40GB for GPU inference. CPU inference is measured with `time.perf_counter` using a single PyTorch thread. Baseline CPU timing is recorded for the iterative FREEGS solver using the same solver configuration as in data generation. Median and p95 statistics are reported over 500 test evaluations. The resulting timing comparison is therefore a comparison

TABLE I. Field-level accuracy versus training set size. Values are averaged over three random initializations. Relative L^2 errors are computed in normalized units and reported as percentages. Physical RMSE is reported in units of Wb after inverse normalization.

N_{train}	rel L^2 (%)	std (%)	RMSE _{norm}	RMSE _{phys} (Wb)
500	0.286	0.034	3.13×10^{-3}	8.53×10^{-5}
1000	0.182	0.002	1.97×10^{-3}	5.33×10^{-5}
2000	0.109	0.004	1.18×10^{-3}	3.16×10^{-5}
5000	0.061	0.006	6.59×10^{-4}	1.79×10^{-5}

against FREEGS as configured in these experiments; optimized production equilibrium solvers or warm-started iterative solves could change the absolute speedup factors.

To quantify physics consistency beyond agreement with FREEGS output, we evaluate an external finite-difference Grad-Shafranov residual diagnostic on the predicted fields. For each predicted field, we first inverse-normalize ψ_{pred} and then compute $\Delta^* \psi_{\text{pred}}$ numerically using second-order central finite differences on the 65×65 grid, where $\Delta^* = R \partial_R (R^{-1} \partial_R) + \partial_Z^2$. The ground-truth right-hand side,

$$\text{RHS}_{\text{true}} = -\mu_0 R^2 \frac{dp}{d\psi} - F \frac{dF}{d\psi}, \quad (10)$$

is loaded directly from the pre-computed `dpdpsi` and `FdFdpsi` fields stored in the dataset, evaluated on the corresponding ground-truth equilibrium. The per-sample normalized residual is

$$\mathcal{R}_i = \frac{\|\Delta^* \psi_{\text{pred}} - \text{RHS}_{\text{true}}\|_{2,\Omega_i}}{\|\text{RHS}_{\text{true}}\|_{2,\Omega_i}}, \quad (11)$$

where $\|\cdot\|_{2,\Omega_i}$ denotes the L^2 norm restricted to interior grid points inside the plasma mask Ω_i . The same quantity is computed for the ground-truth FREEGS fields to establish the finite-difference baseline. The absolute value of this diagnostic should therefore not be interpreted as a solver-internal residual norm; it is a comparative finite-difference diagnostic applied consistently to both predicted and reference fields.

III. RESULTS

A. Field accuracy and data scaling

The most global quantitative trend in the study is the monotonic reduction in test error with increasing training data. Table I reports the test relative L^2 error and physical RMSE for each N_{train} , averaged over three random initializations.

Figure 3 shows the same trend on a log-log scale. A power-law fit to the mean errors yields

$$\epsilon_{\text{rel}} \propto N^{-0.68}, \quad (12)$$

which is steeper than the $N^{-0.5}$ behavior expected from random sampling alone. Fit over four training sizes, this empirical power-law is suggestive rather than definitive – it is consistent with the model exploiting the regularity of the constrained

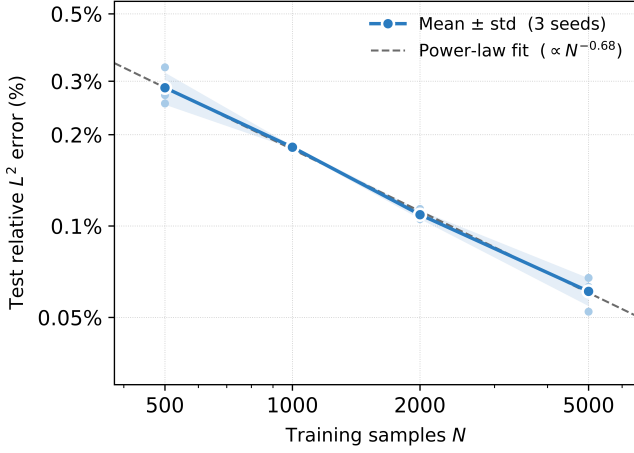


FIG. 3. Test relative L^2 error versus number of training samples on a log-log scale. Filled circles denote means over three initialization seeds, the shaded band spans ± 1 standard deviation and the small open markers show individual seed results. The dashed line is a power-law fit $\epsilon \propto N^{-0.68}$ over the tested range. The fitted trend is consistent with the model exploiting the regularity of the constrained equilibrium family, but is not intended as a definitive sample-complexity law.

equilibrium family within the tested configuration space, but we do not claim it as a precise characterization of the operator's sample complexity. Across the full range from 500 to 5000 training samples, the mean test error drops by a factor of 4.7. By $N = 1000$, the seed-to-seed variance is already very small (std = 0.002%), indicating a stable training regime.

The best-performing model ($N = 5000$, seed 3, best validation epoch 310) achieves a mean relative L^2 error of 0.052% and a median of 0.046% on the 500-sample test dataset. The corresponding physical RMSE is 1.54×10^{-5} Wb. More than 95% of test samples fall below 0.12% relative error, so the reported mean is representative of the bulk of the distribution rather than being driven by a few favorable cases. The full distributions of field-level and geometry-aware errors are shown in Fig. 6.

B. Geometry-aware validation

Field norms alone are not sufficient for diverted equilibria. Additionally, a key question is whether the learned flux preserves the separatrix and the associated critical-point structure. We identify the X-points as saddle points satisfying $\nabla\psi = 0$, while the separatrix is defined by $\psi = \psi_{\text{bdry}}$, where $\psi_{\text{bdry}} = \frac{1}{2}(\psi_X^{\text{lo}} + \psi_X^{\text{up}})$. The geometry-aware metrics for the best $N = 5000$ model across all 500 test equilibria are summarized in Table II.

The separatrix is reproduced with a mean closest-point deviation of 0.072 cm and a median Hausdorff distance of 0.967 cm, while the enclosed area error remains below 0.5% on average. The mean Hausdorff distance is larger (2.128 cm) because it is sensitive to a small subset of cusp-region outliers

TABLE II. Geometry-aware metrics for the best model ($N = 5000$, seed 3) over 500 test equilibria. All distance metrics are in cm. `find_critical` returns two distinct X-points on all 500 test samples; 5.8% of samples (29/500) lie beyond the Tukey fence ($Q_3 + 1.5 \times \text{IQR}$) for at least one X-point. All samples are included in the reported statistics.

Metric	Mean	Median	P95
Separatrix mean error (cm)	0.072	0.059	0.147
Separatrix Hausdorff (cm)	2.128	0.967	4.693
Separatrix area rel. error (%)	0.465	0.343	0.992
X-point lower (cm)	0.161	0.136	0.350
X-point upper (cm)	0.112	0.097	0.236
O-point (cm)	0.031	0.023	0.079
$ \Delta\psi_{\text{bdry}} $ (Wb)	9.82×10^{-6}	—	7.00×10^{-5}

near the X-point pinch, as discussed below. Both X-points are recovered to within approximately 0.1–0.2 cm on average, with the upper null slightly more accurate than the lower one. This residual gap is examined further in Sec. IV A. The magnetic O-point is localized to within 0.031 cm (about 0.3 mm), and the mean boundary-flux error is 9.82×10^{-6} Wb.

Figure 4 shows a near-median test case in which the predicted and reference contour families almost overlap. The near-perfect overlap of the flux contours across the domain illustrates that the low relative L^2 error corresponds to a geometrically faithful equilibrium, while the remaining visible discrepancies are localized near the X-points.

C. Error structure and separatrix ensemble behavior

Figure 5 presents three representative test equilibria: a low-error case, a near-median case, and a near-p95 case. Across all three examples, the global flux-surface structure is reproduced faithfully. The predicted equilibria preserve the separatrix topology and X-point geometry, and no spurious islands or discontinuities are observed.

In addition, it is observed that the error field is smooth and spatially localized. It concentrates near the high-curvature regions around the X-points and along the separatrix, where small perturbations in ψ have the largest geometric effect. Core-region errors remain uniformly small. Even for near-p95 samples, the dominant discrepancy is a modest local shift in flux magnitude rather than a structural failure of the reconstructed equilibrium.

Figure 6 shows the distributions of field-level and geometry-aware errors over all 500 held-out test equilibria. The relative L^2 error distribution is unimodal and right-skewed, with mean 0.052% and median 0.046%, confirming that the best-model performance is typical of the test set rather than exceptional.

The separatrix mean-error distribution is similarly concentrated, with most samples below 0.10 cm. By contrast, the Hausdorff distribution is bimodal: a dominant peak lies below 1 cm, while a secondary peak near 4–5 cm comes from the small subset of cases in which the X-point pinch geometry produces a large local deviation. `find_critical` succeeds

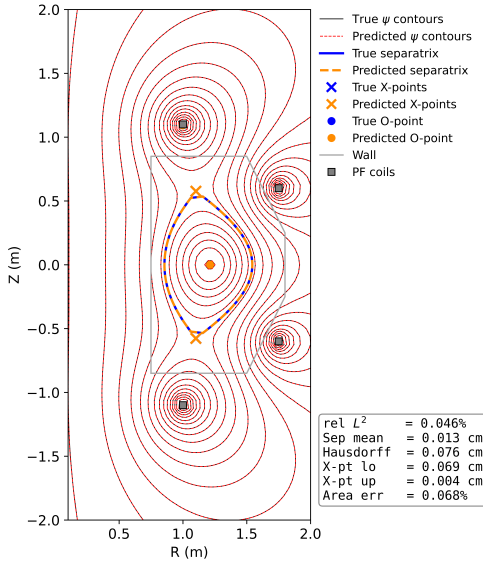


FIG. 4. Poloidal flux contour overlay for a near-median test equilibrium (rel. $L^2 = 0.046\%$, $P_{\text{axis}} = 874$ Pa, $I_p = 1.53 \times 10^5$ A, $f_{\text{vac}} = 2.11$). Solid black lines are ground-truth ψ contours and dashed red lines are FNO predictions on identical iso-flux levels. The true separatrix ($\psi = \psi_{\text{bdry}}$) is shown in dark blue (solid) and the predicted separatrix in orange (dashed). The true and predicted X-points are marked with blue and orange crosses, and the O-points with filled circles of the same colors, respectively. The gray polygon is the wall boundary and the gray squares are PF coil positions. The metrics box reports field-level and geometry-aware errors for this single sample, not ensemble statistics – ensemble means and medians are given in Table II. The residual discrepancies near the X-points are consistent with the geometric amplification of small flux errors at saddle points of ψ .

on all 500 test predictions, so the tail is geometric rather than a detection failure. The X-point histogram also reveals a systematic asymmetry, with the upper X-point recovered more accurately than the lower one (mean 0.112 vs 0.161 cm). The origin of this gap is characterized further in Sec. IV A. O-point errors are tightly concentrated below 0.05 cm for the great majority of samples.

Unified characterization of the tail. We verified that the elevated X-point error in the upper tail is not a distinct failure mode by comparing the 29 Tukey-outlier samples against the 471 inliers. The outlier group exhibits relative L^2 error of 0.075% versus 0.050% for inliers ($1.50\times$ ratio), O-point error of 0.048 cm versus 0.029 cm ($1.63\times$), and separatrix mean error of 0.092 cm versus 0.071 cm ($1.30\times$). All three metrics degrade in concert, with the strongest statistical association seen for the field-level error itself (point-biserial correlation $r = 0.26$, $p = 3 \times 10^{-9}$). The heavy Hausdorff tail is therefore a single underlying phenomenon – samples with slightly lower field-level accuracy produce proportionally larger geometric error, amplified most strongly at the X-point saddle, where local contour location is first-order sensitive to ψ .

Figure 7 overlays all 500 true and predicted separatrices from the held-out test set. Along the main arc, the two bundles are nearly indistinguishable. Visible spread is confined almost

TABLE III. Inference latency and speedup over all 500 test equilibria at batch size 1. GPU measurements use synchronized CUDA events on an NVIDIA A100 SXM4-40GB. CPU measurements use a single PyTorch thread. FREEGS baseline uses Picard iteration with the same solver configuration as data generation. All 500 samples converged successfully.

Method	Median (ms)	p95 (ms)	p95 (median)	Speedup (median)
FNO (GPU)	2.765	2.803	1.01	$\sim 640\times$
FNO (CPU)	25.587	25.810	1.01	$\sim 69\times$
FREEGS (CPU)	1768	2002	1.13	—

entirely to the zoomed regions around the upper and lower X-points, where the separatrix pinches to a cusp and small field errors are geometrically amplified.

This ensemble view clarifies the interpretation of the geometry metrics. In particular, the elevated mean Hausdorff distance is not evidence of a systematic global shape mismatch; rather, it is driven by localized cusp errors in a minority of samples. That behavior is consistent with the smooth error fields in Fig. 5 and with the concentrated separatrix area and mean-distance statistics in Table II.

D. Computational performance and finite-difference GS residual

Inference latency is evaluated at batch size 1 using two timing methods: synchronized CUDA events on an NVIDIA A100 SXM4-40GB for GPU inference, and `time.perf_counter` for CPU inference (one PyTorch thread). FREEGS baseline timing is recorded on the same CPU using the identical solver configuration used during data generation: Picard iteration, $\gamma = 10^{-12}$, iso-flux constraints to a fixed outboard midplane reference, and a maximum of 50 iterations. All 500 test equilibria converged successfully in the FREEGS baseline run. We note that FREEGS is an open-source Python research solver rather than a production real-time equilibrium code. Comparisons against hand-optimized C or Fortran free-boundary solvers, or against FREEGS with warm-started initial guesses, would likely narrow the absolute speedup while leaving the qualitative contrast – second-scale iterative computation versus millisecond-scale deterministic inference – unchanged. Table III summarizes latency statistics and speedup factors.

The FNO surrogate reduces equilibrium evaluation time from second-scale iterative computation to millisecond-scale inference. GPU and CPU speedups relative to FREEGS as configured in these experiments are $\sim 640\times$ and $\sim 69\times$ at the median, respectively. The CPU result is particularly relevant for deployment contexts where GPU hardware is unavailable or where deterministic execution guarantees are required without GPU scheduling overhead.

Equally important is the near-deterministic evaluation time of both FNO variants. The p95/median ratio is 1.01 for FNO (GPU) and FNO (CPU), compared to 1.13 for FREEGS,

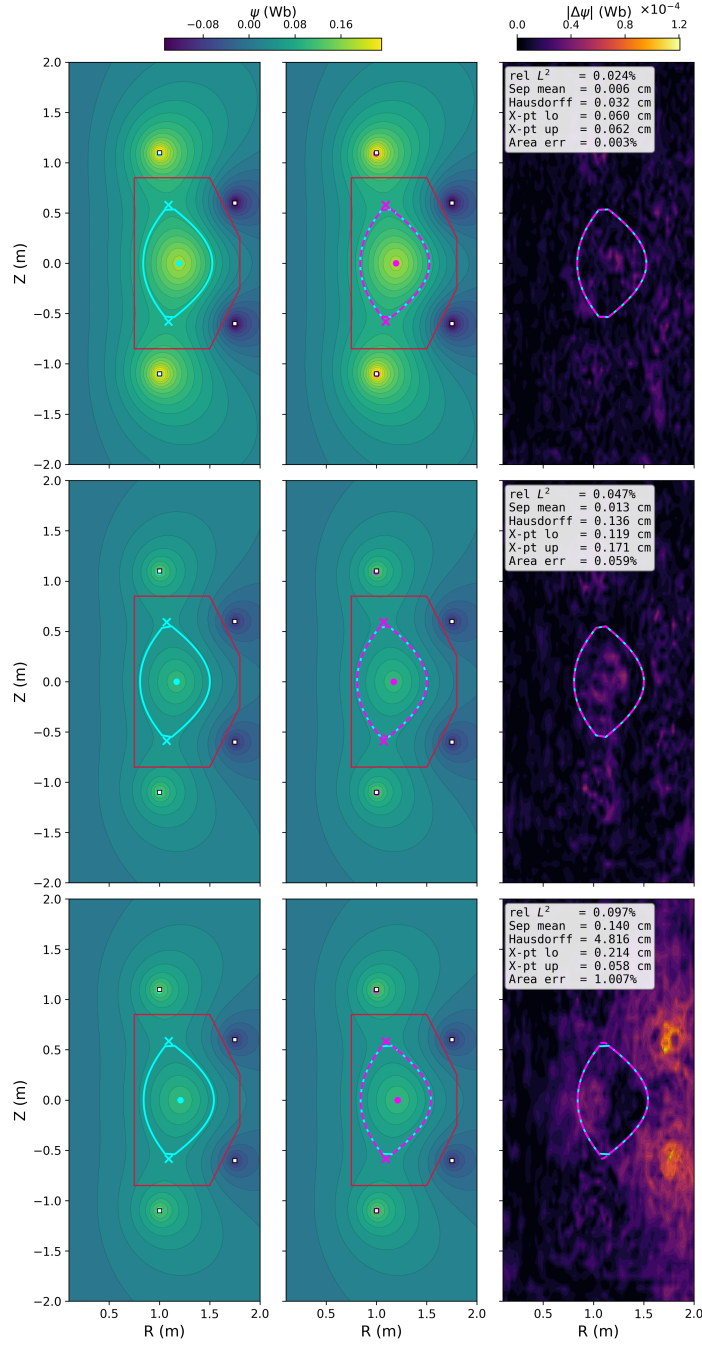


FIG. 5. Poloidal flux field reconstructions for three representative test equilibria selected by relative L^2 error: low error (top row, $\epsilon_{\text{rel}} = 0.024\%$), near-median error (middle row, $\epsilon_{\text{rel}} = 0.047\%$), and near-p95 error (bottom row, $\epsilon_{\text{rel}} = 0.097\%$). Each row shows, from left to right: the ground-truth $\psi(R, Z)$, the FNO prediction, and the absolute pointwise error $|\psi_{\text{pred}} - \psi_{\text{true}}|$. The two field panels in each row share a common color scale (top colorbar), while the error panel uses a separate scale to resolve spatial error structure across the full dynamic range. In the true ψ panels, the cyan contour marks the ground-truth separatrix. In the predicted and error panels, the true separatrix is shown in cyan (solid) and the predicted separatrix in magenta (dashed). The gray polygon denotes the wall boundary and open squares mark PF coil positions. Pointwise error is spatially concentrated near the X-points and along the separatrix, with core-region errors remaining uniformly small across all three cases. Per-panel geometry-aware metrics are reported in the inset boxes.

whose convergence latency varies with the difficulty of each individual equilibrium. The standard deviation of FREEGS solve times (92 ms) is comparable to the entire range of FNO GPU latencies (2.70–4.76 ms across all 500 samples). From

a control-theoretic perspective, predictable evaluation latency is as important as average speed for embedding equilibrium calls inside real-time supervisory loops. Figure 8 shows the empirical CDFs (Cumulative Distribution Function) of evalu-

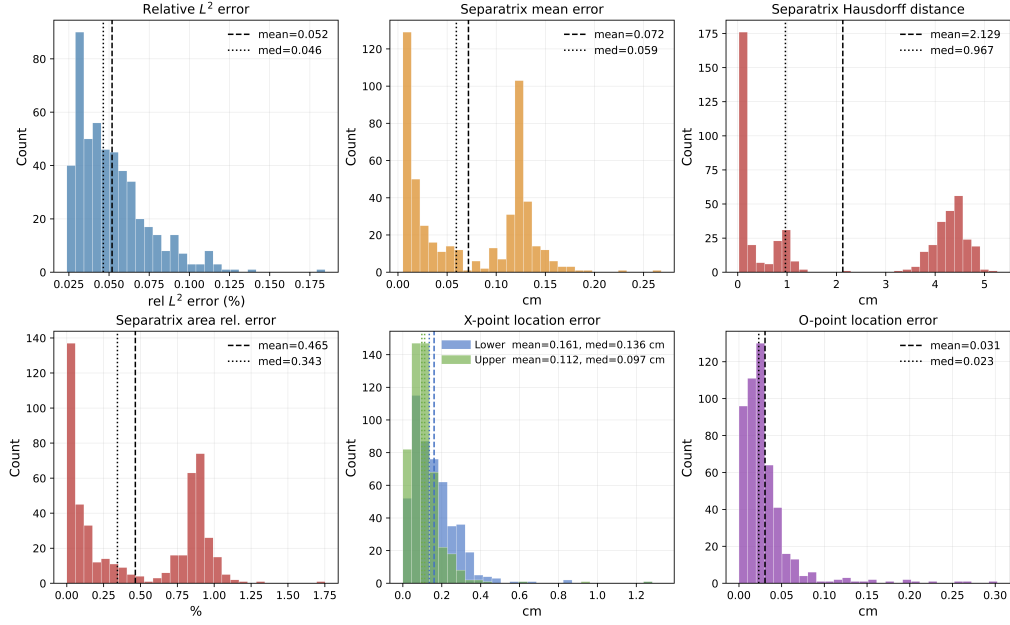


FIG. 6. Distributions of field-level and geometry-aware errors over all 500 test equilibria for the best model ($N = 5000$, seed 3). Top row, left to right: relative L^2 error, separatrix mean closest-point deviation, and separatrix Hausdorff distance. Bottom row: separatrix enclosed area relative error, X-point location errors for the lower (blue) and upper (green) X-points shown as overlapping histograms, and magnetic O-point location error. Dashed and dotted vertical lines mark the mean and median of each distribution, respectively. The heavy right tail in the Hausdorff distribution is driven by a small subset of samples where large deviations arise near the X-point pinch geometry. It does not reflect systematic shape misrepresentation across the ensemble; `find_critical` succeeds on all 500 test predictions, and the Tukey-outlier samples correspond to the upper tail of the field-level error distribution amplified geometrically at the saddle (see Sec. III C). The upper X-point is recovered with systematically lower error than the lower X-point (mean 0.112 vs 0.161 cm).

ation latency for all three methods over the 500 test samples.

Beyond agreement with FREEGS output, we evaluate whether the FNO-predicted fields are physically consistent according to the external finite-difference Grad–Shafranov residual diagnostic defined in Sec. II D. For each of the 500 test predictions, we compute $\Delta^* \psi_{\text{pred}}$ by second-order central finite differences and compare it to the ground-truth right-hand side $\text{RHS} = -\mu_0 R^2 dp/d\psi - F dF/d\psi$, evaluated inside the plasma mask.

The FNO achieves a mean normalized residual of $\mathcal{R} = 2.29$ (median 2.28, p95 2.40), compared to a baseline of 2.29 ± 0.06 obtained by applying the same finite-difference scheme to the ground-truth FREEGS fields. The two values are indistinguishable: the ratio of FNO to ground-truth normalized residual is 0.998. The non-zero baseline reflects the finite-difference truncation error on the 65×65 grid and a slight inconsistency between FREEGS’s internal discretization and our external second-order evaluation – both sources affect predicted and true fields under this diagnostic. Thus, within the tested configuration space, the surrogate satisfies the finite-difference GS residual diagnostic at the same level as the ground-truth FREEGS fields. The absolute value of $\mathcal{R}$ should not be interpreted as a solver-internal residual norm, but as a comparative diagnostic applied consistently to both predicted and reference fields.

IV. DISCUSSION

A. Implications and interpretation

The main result of this study is not merely a reduction in field error, but a change in how constrained free-boundary equilibria can be computed in practice. Once trained, the neural operator replaces the conventional iterative nonlinear Picard scheme with a single deterministic forward evaluation. This is significant because it makes the method directly applicable to workflows in which the equilibrium solve is embedded within a larger outer loop – including model predictive control, rapid scenario exploration, uncertainty quantification, and digital-twin architectures coupling equilibrium, actuator, and transport models. Integration into any of these specific workflows would require end-to-end validation, which we leave to future work.

Experimental real-time reconstruction codes are typically optimized for device-specific diagnostic inversion and rely on temporal coherence between consecutive time slices^{6,7}. The surrogate presented here operates on a different principle: it maps directly from the operating point and prescribed X-point geometry to the flux field, without requiring warm starts or sequential continuity between time slices. In this respect, it should be interpreted as a surrogate for a constrained free-boundary design problem, rather than as an unconstrained actuator-to-equilibrium map. This distinction is what makes it

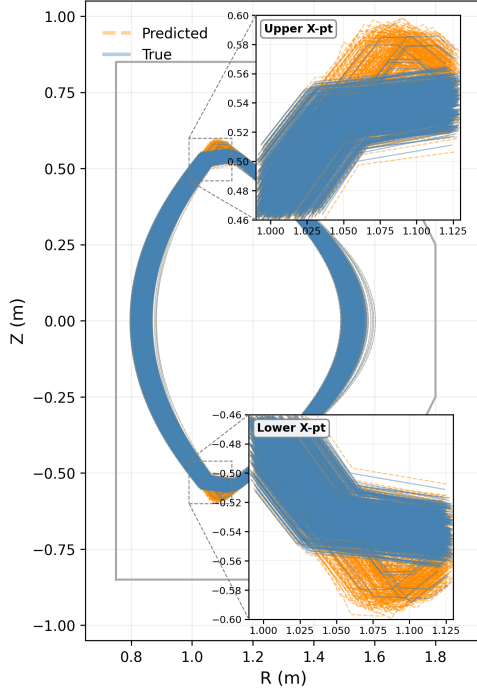


FIG. 7. Overlay of all 500 true (blue, solid) and predicted (orange, dashed) separatrixes from the held-out test set. True and predicted separatrixes are in close agreement across the full test distribution, with the predicted bundle nearly indistinguishable from the true bundle along the main arc. Insets show zoomed views of the upper and lower X-point regions, where localized discrepancies are visible as the separatrix pinches to a cusp. These deviations are responsible for the elevated mean Hausdorff distance relative to the median (2.128 vs 0.967 cm) and reflect the geometric amplification of small flux errors near saddle points of ψ , rather than systematic shape misrepresentation across the separatrix. The wall boundary is shown in gray. PF coil positions are omitted for clarity.

well suited to simulation-based design and optimization tasks, where equilibria at arbitrary operating points and target geometries must be evaluated rapidly and independently of one another.

Field-level relative errors on the order of 5×10^{-4} are already sufficiently small that geometric quantities become the decisive test of performance. It is found that the model recovers the separatrix with a mean closest-point deviation of 0.072 cm and a median Hausdorff distance of 0.967 cm. The X-points are located to within 0.112 cm (upper) and 0.161 cm (lower) on average, and the O-point to within 0.031 cm. Figs. 4–7 show that the residual discrepancies concentrate near the X-point pinch, where small perturbations in ψ produce the largest geometric shifts. Importantly, no test case exhibits a topological failure, such as a spurious flux island or a grossly misplaced separatrix.

The persistent gap between upper and lower X-point accuracy (0.112 vs 0.161 cm) is notable given that the data-generation pipeline is exactly up–down symmetric by construction. We tested two candidate sources of this asymmetry directly on all 500 test samples. Applying `find_critical` to

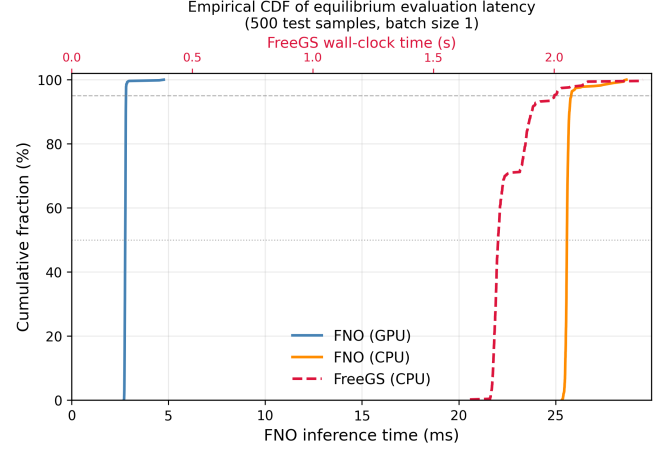


FIG. 8. Empirical cumulative distribution of equilibrium evaluation latency over 500 held-out test samples at batch size 1. The FNO surrogate on GPU (blue) and CPU (orange) exhibit near-deterministic millisecond-scale latency (p95/median = 1.01 for both). The FREEGS iterative solver (red dashed, top axis in seconds) operates at second-scale cost with significant convergence-dependent spread (p95/median = 1.13, std = 92 ms). GPU and CPU FNO speedups are $\sim 640\times$ and $\sim 69\times$ at the median, respectively, relative to FREEGS as configured here.

the ground-truth FREEGS fields and comparing to the nominal symmetric targets gives mean errors of 2.0652 cm and 2.0651 cm for the lower and upper nulls, respectively – a gap of 2×10^{-4} cm, consistent with numerical noise. Applying `find_critical` to each field and to its Z-reflected counterpart gives X-point locations that agree to machine precision. It follows that neither the solver nor the critical-point detector contributes to the observed asymmetry. The residual 0.049 cm gap is most plausibly attributed to a small sensitivity to input-coordinate conventions in a model trained without up–down data augmentation. We note that the gap lies well below the Tukey-outlier thresholds (0.43 cm for the lower null, 0.25 cm for the upper) and does not affect any reported conclusion. Enforcing equivariance through training-time Z-flip augmentation, or through an equivariant architectural prior, would likely eliminate this residual asymmetry.

The empirical $N^{-0.68}$ power-law observed over the tested range is steeper than the $N^{-0.5}$ rate expected from random sampling of a generic high-dimensional function. Since this trend is fit over only four training sizes, it should be regarded as suggestive rather than definitive. It is consistent with the parametric equilibrium family lying on a relatively low-dimensional effective manifold: the scalar operating parameters and X-point coordinates define a smooth deformation of a single double-null topology, and the GS solution operator inherits this smoothness. Whether this behavior persists across broader topology families or machine geometries remains to be tested in future work.

The spatial error structure is also physically interpretable. The pointwise error is found to concentrate near the X-point pinch geometry and along the separatrix, while errors in the core region remain small throughout (Fig. 5). Near an X-

point, the flux gradient vanishes, so that a small absolute error in ψ translates into a comparatively large displacement of the separatrix contour. This geometric amplification effect accounts for both the elevated mean Hausdorff distance relative to the median (2.128 versus 0.967 cm) and the bimodal tail seen in Fig. 6, which reflects a minority of geometrically sensitive cases rather than a systematic misrepresentation of shape.

B. Scope and limitations

This work is deliberately scoped to a single machine geometry and a single topology, and the results should be interpreted accordingly. All training and test data are drawn from a single wall geometry and a single controlled double-null equilibrium family. Accordingly, the results demonstrate strong in-distribution generalization within that family, and should not be taken to imply cross-device or cross-topology transfer. The operator is conditioned on prescribed X-point coordinates, which is natural for design and control workflows in which the divertor geometry is specified, but it does not address diagnostic reconstruction problems in which the geometry must instead be inferred from measurements.

The present model also does not constitute a general actuator-level forward solver. In an actuator-conditioned formulation, the coil currents, vessel currents, and profile information determine the equilibrium, with the X-point locations obtained as outputs. Here, by contrast, the X-point coordinates form part of the input specification. This is a useful formulation for constrained scenario design and target-geometry studies, but it should not be interpreted as a replacement for actuator-to-equilibrium modeling.

All experiments were performed on a fixed 65×65 grid; transfer to other resolutions was not tested. The surrogate models static GS equilibria and is not yet coupled to transport evolution, actuator dynamics, or stability calculations. The speedup factors reported here are measured relative to FREEGS as configured in the present experiments. Optimized production equilibrium solvers, hand-tuned C or Fortran implementations, or warm-started iterative solves may reduce the absolute speedup factors obtained; however, the qualitative contrast between iterative, sample-dependent latency and deterministic, millisecond-scale inference is expected to remain. Finally, experimental deployment would require hardened software integration and careful validation under distribution shift, both of which lie beyond the scope of the present work.

C. Future extensions and broader perspective

A natural next step is the construction of coupled equilibrium–transport operator surrogates. In integrated modeling, transport solvers repeatedly call equilibrium routines to update the magnetic geometry, and the repeated GS solve is often the dominant cost¹¹. A paired surrogate could thus

enable fast self-consistent updates within optimization and digital-twin loops.

A second direction is to broaden the training distribution across magnetic topologies, including lower and upper single-null, limiter, and snowflake configurations, and eventually across machine geometries. This would be valuable for cross-device studies and for controller design on next-step devices such as ITER⁴⁷ and SPARC⁴⁸. Multi-fidelity training strategies could combine large volumes of lower-cost data with smaller amounts of high-fidelity constrained free-boundary solutions.

We also plan to test the resolution-transfer properties of the FNO explicitly. A single trained operator that evaluates consistently across grids would interface naturally with multi-fidelity transport, stability, and control workflows. In addition, physics-aware losses or auxiliary supervision on derived equilibrium quantities – finite-difference GS residual diagnostics, safety-factor proxies, or shape descriptors – may help suppress the localized cusp errors that dominate the Hausdorff tail, and would make the surrogate easier to couple to downstream codes. A simple ablation of the X-point conditioning would also be a useful direction for future work, since it would quantify how much of the geometric fidelity is attributable to explicitly prescribing the divertor geometry.

This work illustrates a general paradigm for fusion AI: repeated nonlinear PDE solves over related operating points can be recast as operator-learning problems, and once the operator has been learned, the cost of high-fidelity simulation can be amortized without discarding physically meaningful outputs. Free-boundary equilibrium is a compelling first target for this paradigm, since both geometry and latency matter for the use cases that motivate the approach. The $\sim 640 \times$ GPU and $\sim 69 \times$ CPU speedups reported here – measured relative to FREEGS as configured in the present experiments – combined with near-deterministic evaluation time (p95/median = 1.01), demonstrate that FNO surrogates can play a practical role in accelerating equilibrium evaluation in modern fusion workflows, including settings in which dedicated GPU hardware is unavailable.

V. SUMMARY AND OUTLOOK

We have demonstrated millisecond-scale neural-operator surrogate modeling for constrained double-null free-boundary Grad–Shafranov equilibria. The FNO learns a constrained forward map from spatial coordinates, scalar operating parameters, and prescribed X-point geometry to the full poloidal-flux field on a 65×65 grid, trained on FREEGS data for a single fixed machine geometry and prescribed topology with 100% solver acceptance rate across all 6000 candidate equilibria.

On 500 test equilibria, the best model achieves a mean relative L^2 error of 0.052% and a physical RMSE of 1.54×10^{-5} Wb. It recovers the separatrix to within 0.072 cm mean closest-point deviation, both X-points to sub-0.2 cm, and the O-point to 0.031 cm. Test error follows an empirical $N^{-0.68}$ power law over $N_{\text{train}} \in \{500, 1000, 2000, 5000\}$, suggesting systematic improvement with training-set size over

the tested regime. The predicted fields satisfy the external finite-difference GS residual diagnostic at the same level as the ground-truth FREEGS fields, with a mean normalized residual of 2.29, indistinguishable from the 2.29 ± 0.06 baseline obtained using the same finite-difference scheme.

GPU inference takes 2.77 ms per equilibrium and CPU inference takes 25.6 ms, corresponding to speedups of $\sim 640\times$ and $\sim 69\times$ relative to FREEGS as configured here. Both FNO variants show near-deterministic latency (p95/median = 1.01), compared to 1.13 for the iterative solver. These results demonstrate that neural operators can deliver accurate, geometrically faithful, and finite-difference physics-consistent equilibrium surrogates for control-oriented workflows, integrated modeling, and large parametric studies in magnetic-confinement fusion, within a prescribed topology and machine geometry. Future directions include extension to multiple topologies and machine geometries, coupled equilibrium-transport surrogates, resolution-transfer studies, and physics-aware training losses.

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DATA AND CODE AVAILABILITY

The dataset generation script, training code, and evaluation notebook used in this work will be made available upon reasonable request to the corresponding author. The FREEGS solver used for data generation is publicly available at <https://github.com/freegs-plasma/freegs>.

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